Lecture 10

Errors and the first codes

Objectives. *Lecture 9 closed the second answer with gates that never err. This lecture opens the third block, protection, and asks:* can a qubit be protected without copying it? *Noise is a gate on the system and an environment that is then discarded (Theorem 10.2). The bit-flip code corrects one flipped qubit without copying (Theorem 10.4). The main result: Shor's nine-qubit code corrects every one-qubit error, as its syndrome measurement collapses a continuous error onto a discrete one (Theorem 10.6).*

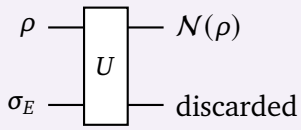
By the end of this lecture you will be able to:

- Derive the Kraus operators of noise from a gate on the system and an environment in a fixed state, and show that a correlated start gives no map of the system's state alone.
- Prove that the bit-flip code corrects one flipped qubit, compute its failure probability $3p^2 - 2p^3$ under independent flips, and show that under independent phase flips with $0 < p < \frac{1}{2}$ it does worse than no code.
- Prove that Shor's code corrects every noise whose Kraus operators each act on one qubit, and say how its syndrome measurement makes a continuous error discrete.

10.1 Noise as an environment

Every gate so far did what its matrix says. A real qubit also couples to its surroundings, the *environment*, which no one controls or reads.

Definition 10.1 (Noise from an environment)**.** Let S be a register, the *system*, and E a further register, the *environment*. The system starts in a density operator ρ, and the environment in a fixed density operator σ_E, which does not depend on ρ: together, in $\rho \otimes \sigma_E$. A gate U on SE acts, and E is discarded, leaving S in its reduced state (Theorem 3.9):



The *noise* is the map $\mathcal{N}(\rho) = \mathrm{Tr}_E\big(U(\rho \otimes \sigma_E)U^\dagger\big)$.

Theorem 10.2 (Kraus operators)**.** *Write $\sigma_E = \sum_j \lambda_j \left|e_j\right\rangle \left\langle e_j\right|$, with $\{e_j\}$ an orthonormal basis of the environment and $\lambda_j \geq 0$, and let $\iota_k = \mathbb{1} \otimes \left|e_k\right\rangle$. Then for every density operator ρ of S*

$$\mathcal{N}(\rho) = \textstyle\sum_{k,j} K_{kj} \rho K_{kj}^\dagger,$$

where the Kraus operators $K_{kj} = \sqrt{\lambda_j}\, \iota_k^\dagger U \iota_j$ *do not depend on ρ, and $\sum_{k,j} K_{kj}^\dagger K_{kj} = \mathbb{1}$.*

Proof. By the interchange law, $\iota_j \rho \iota_j^\dagger = \rho \otimes |e_j\rangle\langle e_j|$, so $\rho \otimes \sigma_E = \sum_j \lambda_j \iota_j \rho \iota_j^\dagger$. By (3.5) in the basis $\{e_k\}$,

$$\begin{aligned}\mathcal{N}(\rho) &= \textstyle\sum_k \iota_k^\dagger U(\rho \otimes \sigma_E)U^\dagger \iota_k \\ &= \textstyle\sum_{k,j} \lambda_j \, \iota_k^\dagger U \iota_j \, \rho \, \iota_j^\dagger U^\dagger \iota_k,\end{aligned}$$

which is the sum claimed. As $\sum_k \iota_k \iota_k^\dagger = \mathbb{1}$, $U^\dagger U = \mathbb{1}$, $\iota_j^\dagger \iota_j = \mathbb{1}$ and $\sum_j \lambda_j = \operatorname{Tr} \sigma_E = 1$,

$$\begin{aligned}\textstyle\sum_{k,j} K_{kj}^\dagger K_{kj} &= \textstyle\sum_j \lambda_j \, \iota_j^\dagger \iota_j \\ &= \mathbb{1}.\end{aligned}$$

■

A mixed environment needs no purification: each eigenvector e_j gives Kraus operators of its own (Nielsen and Chuang, p. 360, purify σ_E and call the K *operation elements*). If U acts on E and on some wires of S only, so does every K_{kj}, by the interchange law.

Example 10.3 (The bit-flip channel)**.** Let the environment be one qubit, a coin in $\sigma_E = (1-p)\,|0\rangle\langle 0| + p\,|1\rangle\langle 1|$, and let $U = \mathbb{1} \otimes |0\rangle\langle 0| + X \otimes |1\rangle\langle 1|$, the CNOT with the control on E: the coin flips the system's qubit when it shows 1. With $e_k = |k\rangle$, $K_{00} = \sqrt{1-p}\,\mathbb{1}$, $K_{11} = \sqrt{p}\,X$ and $K_{01} = K_{10} = 0$:

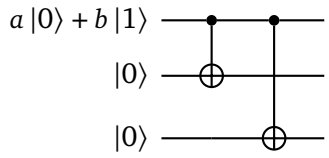
$$\mathcal{N}(\rho) = (1-p)\rho + pX\rho X.$$

On $|0\rangle\langle 0|$ it gives $\operatorname{diag}(1-p, p)$, as Lecture 1's noisy wire does. With Z in place of X it is the *phase-flip channel* $(1-p)\rho + pZ\rho Z$, which leaves $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ alone and takes $|+\rangle\langle +|$ to $(1-p)\,|+\rangle\langle +| + p\,|-\rangle\langle -|$.

The start $\rho \otimes \sigma_E$ is needed. With U as in the example, the starts $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ and $\frac{1}{2}\mathbb{1} \otimes |0\rangle\langle 0|$ both leave S in $\frac{1}{2}\mathbb{1}$; after U, the first leaves it in $|0\rangle\langle 0|$, the second in $\frac{1}{2}\mathbb{1}$. No map of ρ alone gives both. Nielsen and Chuang (p. 358) promise that their formalism covers such starts; their §8.5 (p. 395) shows that it fails even for $\rho \otimes \sigma_E$ with σ_E depending on ρ.

10.2 The bit-flip code

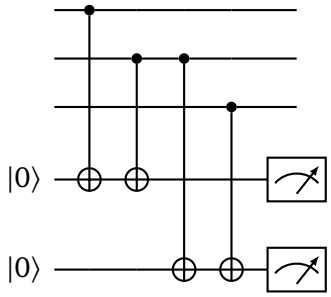
A qubit cannot be copied (Theorem 4.1), as the bit of Example 1.6 was, but its basis can, by two CNOTs (Lecture 4):



gives $a\,|000\rangle + b\,|111\rangle$. Unless $ab = 0$, this is not $\psi \otimes \psi \otimes \psi$, whose amplitude of $|001\rangle$ is $a^2 b$; and each qubit alone is in $|a|^2\,|0\rangle\langle 0| + |b|^2\,|1\rangle\langle 1|$ (Corollary 3.10), without the relative phase of a and b. The *code* is the plane C of $|000\rangle$ and $|111\rangle$.

Let X_i be X on qubit i and $\mathbb{1}$ on the others; Z_i likewise. The error X_1 moves C to the plane of $|100\rangle$ and $|011\rangle$. To locate it without reading a or b, read only the parities $s = (x_1 \oplus x_2,\ x_2 \oplus x_3)$, the *syndrome*: Π_s projects onto the span of the strings x with syndrome s, so Π_{00} onto C and

Π_{10} onto $X_1 C$. The circuit



maps $|x\rangle|00\rangle$ to $|x\rangle|s\rangle$, so $\psi|00\rangle$ to $\sum_s \Pi_s\psi \otimes |s\rangle$: by (3.2), reading the lower wires is the measurement $\{\Pi_s\}$ of the upper ones.

Theorem 10.4 (The bit-flip code). *Let $C_{00} = \mathbb{1}$, $C_{10} = X_1$, $C_{11} = X_2$ and $C_{01} = X_3$. The* recovery *measures the syndrome and applies C_s on the outcome s.*

(i) For $E = \mathbb{1}, X_1, X_2, X_3$ and every $\psi \in C$, the syndrome measurement on $E\psi$ gives $s(E) = 00, 10, 11, 01$ respectively, with certainty; it leaves $E\psi$, and $C_{s(E)}E\psi = \psi$.

(ii) If each qubit of $\psi \in C$ is flipped independently with probability p, the recovery leaves $(1-q)|\psi\rangle\langle\psi| + q\bar{X}|\psi\rangle\langle\psi|\bar{X}$, where $\bar{X} = X \otimes X \otimes X$ and $q = 3p^2 - 2p^3$.

Proof. (i) X_i flips the bit i of 000 and of 111, taking both into the range of $\Pi_{s(E)}$, as $\mathbb{1}$ does. So $\Pi_{s(E)}E\psi = E\psi$, with probability $\|E\psi\|^2 = 1$, and $C_{s(E)}E\psi = \psi$ as $X_i^2 = \mathbb{1}$.

(ii) With one coin per qubit (Example 10.3), the gate takes $|x\rangle|c\rangle$ to $X^c|x\rangle|c\rangle$, with $X^c = X^{c_1} \otimes X^{c_2} \otimes X^{c_3}$, and Theorem 10.2 gives the nonzero Kraus operators $\sqrt{w_c}\,X^c$, $w_c = p^{|c|}(1-p)^{3-|c|}$, for $c \in \{0,1\}^3$ with $|c|$ ones. Averaged over the outcome, the recovery is $\mathcal{R}(\tau) = \sum_s C_s\Pi_s\tau\Pi_s C_s^\dagger$, by (3.3): a term of probability 0 vanishes, as $\Pi_s\tau\Pi_s$ has no negative eigenvalue and trace 0. Now $X^c\psi$ lies in the range of Π_s for s the syndrome of c, and $C_s = X^f$ for the f with $|f| \le 1$ and that syndrome; $c \oplus f$ has syndrome 00, so it is 000 if $|c| \le 1$ and 111 if $|c| \ge 2$. So $\mathcal{R}$ takes $X^c|\psi\rangle\langle\psi|X^c$ to $|\psi\rangle\langle\psi|$ or $\bar{X}|\psi\rangle\langle\psi|\bar{X}$, and the w_c with $|c| \ge 2$ add up to $3p^2(1-p) + p^3 = q$. ∎

The syndrome locates the error and reveals nothing about ψ: its distribution is the same for every $\psi \in C$. The failure probability q is that of Example 1.6's majority, as Lecture 1 promised. As $q - p = -p(1-p)(1-2p)$, the code helps exactly when $0 < p < \frac{1}{2}$. But $q > 0$ for every $p > 0$: failure is suppressed, not removed.

10.3 Phase flips by conjugation

The bit-flip code is blind to phases: $Z_1(a|000\rangle + b|111\rangle)$ is $a|000\rangle - b|111\rangle$, in C, so the syndrome is 00 and nothing is corrected. On C an even number of Z_i acts as $\mathbb{1}$, and an odd number as $\bar{Z} = Z \otimes Z \otimes Z$. Under independent phase flips the recovery leaves $\bar{Z}\psi$ with probability $3p(1-p)^2 + p^3$, and

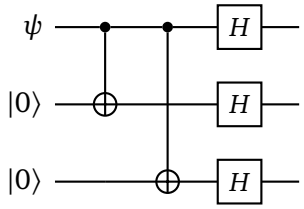
$$3p(1-p)^2 + p^3 - p = 2p(1-p)(1-2p)$$

is positive for $0 < p < \frac{1}{2}$: the code does worse than no code (Watrous, 2025, pp. 380–381).

Proposition 10.5 (Conjugated codes). *Let V be a gate. If the measurement $\{\Pi_s\}$ and the corrections C_s correct an error E on every $\psi \in C$, as in Theorem 10.4(i), then $\{V\Pi_s V^\dagger\}$ and $VC_sV^\dagger$ correct $VEV^\dagger$ on every vector of VC.*

Proof. $\{V\Pi_s V^\dagger\}$ is a projective measurement, and $VEV^\dagger(V\psi) = VE\psi$. By hypothesis $\Pi_{s(E)}E\psi = E\psi$ and $C_{s(E)}E\psi = \psi$; apply V. ∎

With $V = H^{\otimes 3}$, $VX_iV^\dagger = Z_i$, as $HXH = Z$. The *phase-flip code*, the plane of $|{+}{+}{+}\rangle$ and $|{-}{-}{-}\rangle$, encoded by



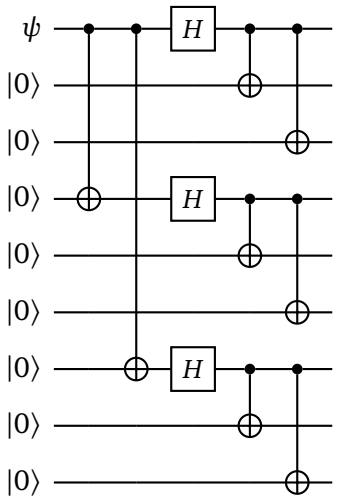
corrects one phase flip by measuring X_1X_2 and X_2X_3: the syndrome circuit between two layers of H (as in Proposition 3.6). It fails as the bit-flip code does, with X and Z exchanged: neither code corrects both.

10.4 Shor's code

Shor's code nests the two: the phase-flip code, with each of its qubits encoded again by the bit-flip code. Its nine qubits form three *blocks*, 1–3, 4–6 and 7–9, and its codewords are

$$|0_L\rangle = 2^{-3/2}\big(|000\rangle + |111\rangle\big)^{\otimes 3},$$
$$|1_L\rangle = 2^{-3/2}\big(|000\rangle - |111\rangle\big)^{\otimes 3}.$$

The circuit



maps $(a\,|0\rangle + b\,|1\rangle)\,|0\cdots0\rangle$ to $a\,|0_L\rangle + b\,|1_L\rangle$: it is the phase-flip encoder on wires 1, 4, 7, and then in each block two CNOTs, which map $|{\pm}\rangle\,|00\rangle$ to $(|000\rangle \pm |111\rangle)/\sqrt{2}$. Let Π_C project onto the code, the plane of $|0_L\rangle$ and $|1_L\rangle$.

The syndrome comes from eight observables that fix both codewords. Six are the blocks' bit-flip syndromes, Z_1Z_2, Z_2Z_3, Z_4Z_5, Z_5Z_6, Z_7Z_8, Z_8Z_9. As $X_1X_2X_3(|000\rangle \pm |111\rangle) = \pm(|000\rangle \pm |111\rangle)$, the other two, $X_1\cdots X_6$ and $X_4\cdots X_9$, compare the signs of blocks 1 and 2, and of blocks 2 and 3. A Z-pair and an X-product share no qubit or two, so any two of the eight commute. Call them $g_1, \ldots, g_8$, and for $s \in \{0,1\}^8$ let

$$\Pi_s = \textstyle\prod_{i=1}^{8} \frac{1}{2}\big(\mathbb{1} + (-1)^{s_i} g_i\big).$$

As $g_i^\dagger = g_i$ and $g_i^2 = \mathbb{1}$, the factors are commuting projectors, so Π_s is one; for $s \neq s'$ some i gives $(\mathbb{1}+g_i)(\mathbb{1}-g_i) = 0$; and $\sum_s \Pi_s = \mathbb{1}$, expanding the product of the sums $\frac{1}{2}(\mathbb{1}+g_i)+\frac{1}{2}(\mathbb{1}-g_i) = \mathbb{1}$. So $\{\Pi_s\}$ is a projective measurement, the *syndrome measurement*. For the bit-flip code, Z_1Z_2 and Z_2Z_3 give the Π_s of §10.2.

If $g_iE = (-1)^{s_i}Eg_i$ for every i, then $g_iE\psi = (-1)^{s_i}E\psi$ for ψ in the code: $E\psi$ lies in the range of $\Pi_{s(E)}$, $s(E) = s$, whose bit s_i is 1 exactly when E anticommutes with g_i. Let $Y_i = iX_iZ_i$. The *28 errors* are $\mathbb{1}$ and X_i, Y_i, Z_i for $i = 1, \dots, 9$. X_i anticommutes with the Z-pairs that contain i: it sets its block's two bits to 10, 11 or 01 as i is the block's first, second or third qubit. Z_i anticommutes with the X-products that contain i: it sets the last two bits to 10, 11 or 01 as i is in block 1, 2 or 3. Y_i does both. So the 28 errors have 22 syndromes; two share one only if they are Z_i and Z_j in one block, and then Z_iZ_j fixes every code vector.

10.5 Every one-qubit error

Theorem 10.6 (Shor's code). *For the syndrome of each of the 28 errors let C_s be that error, with Z_1, Z_4, Z_7 for those of the Z_i of blocks 1, 2, 3, and let $C_s = \mathbb{1}$ for any other s. The recovery measures the syndrome and applies C_s.*

(i) For each of the 28 errors E and every ψ in the code, the syndrome measurement on $E\psi$ gives $s(E)$ with certainty and leaves $E\psi$, and $C_{s(E)}E\psi = \psi$.

(ii) Let $\mathcal{N}(\tau) = \sum_k K_k\tau K_k^\dagger$, with $\sum_k K_k^\dagger K_k = \mathbb{1}$ and each K_k acting on one qubit, which may differ from one k to another. Then the recovery, averaged over its outcome, has $\mathcal{R}(\mathcal{N}(\rho)) = \rho$ for every density operator ρ with $\Pi_C\rho\Pi_C = \rho$.

Proof of (i). By the syndromes above, $E\psi$ lies in the range of $\Pi_{s(E)}$, so the measurement gives $s(E)$ with probability $\|E\psi\|^2 = 1$ and leaves $E\psi$. If $C_{s(E)} = E$, then $C_{s(E)}E = E^2 = \mathbb{1}$; otherwise $E = Z_j$ and $C_{s(E)} = Z_i$ in one block, and $Z_iZ_j\psi = \psi$. ∎

Part (i) corrects 28 errors; part (ii) says that they suffice.

Proof of (ii). With $\sigma_0, \dots, \sigma_3 = \mathbb{1}, X, Y, Z$, every 2×2 matrix is $K = \sum_a c_a\sigma_a$, for $c_a = \frac{1}{2}\operatorname{Tr}(\sigma_aK)$. So a K_k on qubit i is a combination of $\mathbb{1}$, X_i, Y_i and Z_i, which have four different syndromes. By (i), and as the Π_s project onto orthogonal subspaces, $\Pi_sK_k\Pi_C$ is $cE\Pi_C$ for the one term cE of K_k with $s(E) = s$, and 0 if there is none; by (i) again, $C_sE\Pi_C = \Pi_C$. So for every k and s

$$C_s\Pi_sK_k\Pi_C = \lambda_{ks}\Pi_C$$

for a number λ_{ks}. Averaged over the outcome, the recovery is $\mathcal{R}(\tau) = \sum_s C_s\Pi_s\tau\Pi_sC_s^\dagger$, as in Theorem 10.4. As $\rho = \Pi_C\rho\Pi_C$,

$$\mathcal{R}(\mathcal{N}(\rho)) = \textstyle\sum_{k,s} |\lambda_{ks}|^2\,\rho.$$

Both maps keep the trace: $\mathcal{N}$ as $\sum_k K_k^\dagger K_k = \mathbb{1}$, and $\mathcal{R}$ as $\sum_s \Pi_sC_s^\dagger C_s\Pi_s = \sum_s \Pi_s = \mathbb{1}$, each C_s being unitary. So $\sum_{k,s}|\lambda_{ks}|^2 = \operatorname{Tr}\rho = 1$. ∎

The identity $C_s\Pi_sK_k\Pi_C = \lambda_{ks}\Pi_C$ holds as well with $\mathbb{1}$ on further qubits, so a code block entangled with qubits that the noise does not touch is restored too. The K_k need not be unitary, and an environment that touches one qubit gives Kraus operators on that qubit (Theorem 10.2).

Example 10.7 (A small rotation)**.** Let the error be a rotation of qubit 1, $e^{-i\epsilon\, n\cdot\sigma/2} = \cos(\epsilon/2)\mathbb{1} - i\sin(\epsilon/2)\, n\cdot\sigma$, with $n\cdot\sigma = n_xX_1 + n_yY_1 + n_zZ_1$ for a unit vector n: one Kraus operator. On ψ in the code, the syndrome measurement gives the syndrome of $\mathbb{1}$ with probability $\cos^2(\epsilon/2)$, and those of X_1, Y_1, Z_1 with probabilities n_x^2, n_y^2, n_z^2 times $\sin^2(\epsilon/2)$. It leaves ψ, $X_1\psi$, $Y_1\psi$ or $Z_1\psi$, each up to a global phase, and the correction returns ψ up to that phase. For $\epsilon = 0.1$, a whole error occurs with probability $\sin^2(0.05) \approx 0.0025$.

The rotation is continuous, the syndrome discrete. The measurement never finds a fraction of an error: it finds none, or a whole one, and leaves a state that one of the 28 corrections repairs. This is how the syndrome measurement collapses a continuous error onto a discrete one, and why correcting 28 errors protects against a continuum: the proof of (ii) used only that each K_k is a combination of the 28. Lecture 11 finds the exact condition under which a code corrects a set of errors, and proves that its recovery then corrects every linear combination of them.

Protection is not perfection. Two errors can defeat the code: X_1X_2 has the syndrome of X_3, and the correction leaves $X_1X_2X_3$, which fixes $|0_L\rangle$ and reverses the sign of $|1_L\rangle$. And the encoding, the syndrome circuit and the corrections were taken to be exact; Lecture 13 states what survives when they too are noisy.

So a qubit can be protected without copying it: the code spreads it over nine qubits, and the syndrome measurement asks only which error occurred, a question whose answer says nothing about the state.

Summary.

- *Noise:* a gate on the system and an environment in a fixed state, which is then discarded, acts as $\rho \mapsto \sum K\rho K^\dagger$ with $\sum K^\dagger K = \mathbb{1}$ (Theorem 10.2); a correlated start gives no map of ρ alone.
- *The bit-flip code:* two CNOTs encode without copying; the syndrome locates one flip and reveals nothing about the state; under independent flips the code fails with probability $3p^2 - 2p^3$ (Theorem 10.4), and under independent phase flips with $0 < p < \frac{1}{2}$ more often than no code.
- *The main result:* Shor's code corrects every noise whose Kraus operators each act on one qubit; its syndrome measurement collapses a continuous error onto one of 28 (Theorem 10.6).
- *Next:* Lecture 11, when is an error correctable?

References

Nielsen and Chuang, pp. 358–360, 395 and 426–434. Watrous (2018), p. 89. Watrous (2025), pp. 374–384 and 396–400 (little-endian). De Wolf, pp. 184–187. P. W. Shor, Phys. Rev. A **52** (1995), R2493–R2496.